

The Writing Assignment

The writing assignment for the course Academic Writing Skills is a literature review. This document will explain what a literature review is and what the context is of your literature review.

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What is a literature review?

A literature review is a survey of scholarly sources on a specific topic. It provides an overview of current knowledge, allowing you to identify relevant theories, methods, and gaps in the existing research. Writing a literature review involves finding relevant publications (such as books and peer reviewed journal articles), critically analysing them, and explaining what you found. A good literature review does not just summarize sources – it analyses, synthesises, and critically evaluates to give a clear picture of the state of knowledge on the subject.

The main goal of a literature review is to identify scientific gaps or knowledge gaps in a certain area of research. To do this, you will perform several literature searches, and analyse the literature you found. By comparing and contrasting the information, you will get a grasp of the state of the art on the topic you have chosen. Moreover, you can identify knowledge gaps on your topic, which lead to recommendations for future research.

Relevance

With this assignment, you will get a glimpse of being a scientist. While some people may check their social media or read a newspaper in the morning, scientific literature is the main reading material of scientists. They read and review scientific articles in their field almost daily. Knowing the state of the art in their field and knowing what challenges remain will help scientists decide on their next research project. Moreover, you will need these skills for your master's thesis: you will probably start with a review of the existing literature before you can decide on your main research question.

What is the context of your literature review?

You need to write a literature review for your Electrical Engineering peers, so not only the peers from your track. This means that the review needs to be understandable for all master

students of Electrical Engineering. The goal of your literature review is to provide your readers with up to date information of a particular part of your field (Profile Orientation) and that you add something new to this field (Academic Skills part), such as an insight, critique, an advice or a well-motivated suggestion for further research (scientific gap).

Requirements

Your literature review has to meet the following requirements:

- **Genre:** literature review
- **Authors:** writing teams of three students
- **Target audience:** peers (master students Electrical Engineering in general, so not only peers from your track)
- **Goal:** to provide your readers with up to date information of a particular part of your field (the Profile Orientation part) and to add something new to this field (the Academic Skills part), such as an insight, critique, an advice or a suggestion for further research (scientific gap).
- **Length:** 2000 words (+/- 10%), excl. abstract, list of references, list of abbreviations, legends and captions of illustrations (group of 2: same length; group of 4: 2700 words)
- **Sources:** minimum of 9 sources (12 for group of 4, minimal 2 sources should be from 2024 or later).
- **Reference style:** IEEE
- **Language:** the paper is written in correct English
- **Format:** the paper is formatted following the IEEE author guidelines
- **Elements:** the document contains the following elements:
 - Title
 - Authors & Affiliation
 - Student names, student numbers, submission date
 - Footnote on the first page, following this template: “This article was written as part of the course EE4C01 Profile Orientation & Academic Skills of the faculty Electrical Engineering, Mathematics and Computer Science of Delft University of Technology. Total word count: [fill in word count].”
 - Abstract
 - Introduction
 - Method
 - Results
 - Note: it is allowed to use other title(s) for the results section(s)
 - Discussion/conclusion
 - List of references
 - Appendix about the use of generative AI in the form of an Acknowledgement

If you would like to deliberately deviate from these requirements, please confer with your lecturer as soon as possible.

Deadline

See Brightspace - Assignments

Grading

Your lecturer will grade your paper using the Grading sheet Scientific Writing. Before grading your paper, your lecturer will check if your paper meets the killer criteria (minimal requirements to pass). The killer criteria are:

- The document is handed in ultimately at the deadline via Brightspace. Late submissions will be regarded as retake entries.
- The document meets the requirements described above
- The document passes the plagiarism check

If the document does not meet the killer criteria, your lecturer will give the submission a fail and they will ask you to submit an improved version (this will be your retake entry).

How to write this literature review?

1. Form a trio with two other students
2. Choose a topic that is relevant to your field of study. No inspiration? Choose a topic from the list with starting articles on Brightspace.
3. Perform a literature search on your topic. Minimum of 9 academic sources, for example journal articles, conference proceedings and academic textbooks.
4. Introduce the chosen topic and legitimate this choice; scientific relevance and/or social relevance.
5. Compare scientific literature on your chosen topic in an analytical way; what do the authors say about the topic? How do they differ, where do they overlap? Evaluate critically. Make sure that you keep track of your findings by organizing your sources.
6. Give meaning to the comparative analysis: can differences/overlaps be explained by the (diverse) backgrounds of the authors, be it institutional, cultural or otherwise? Motivate this analysis.
7. Conclude your analysis with a clear picture of the state of knowledge on the topic: what is the accepted/debated state-of-the-art and what are the issues currently disputed and/or on the agenda for further research and debate?
8. Translate the insights you gained - this new scientific knowledge - into recommendations for other scientists.